Bayan AI

بيان

A clinical AI system that interprets between doctors and patients in Qatar, reasons about the consultation, and flags what gets missed.

Team QH-129 Qai × Hugging Face Hackathon 01 Doha, 5–7 Nov 2026

Meet Anurag

29, a Nepali construction worker in Doha

Anurag says

“My chest is burning. It’s just *gyastrik*.”

ग्यास्ट्रिक (gyastrik) is Nepali for “gastric”

The doctor hears

A stomach problem, like reflux

It could be

His heart

Language barriers in Qatar

65%

of HMC ambulance staff face language barriers at work

71%

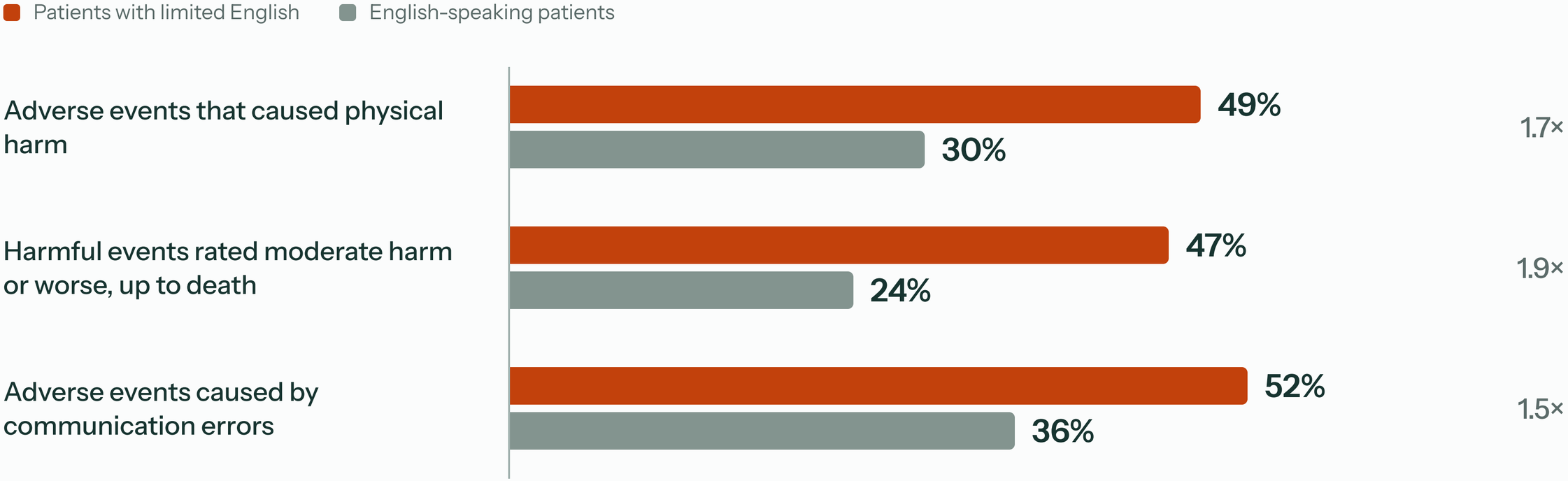
are comfortable letting family members interpret

9

languages with face-to-face interpreters at Sidra; Nepali, Malayalam, Bengali and Tagalog are not among them

Adverse events and language

Many patients in Qatar don’t share a language with their doctor. US hospitals that measured the effect found this.



Bayan is built to close this gap: it interprets, reasons about the consultation and flags what gets missed.

What Bayan does

It listens to both sides of the consultation.

Interprets

37 patient languages. The doctor speaks English or Arabic.

Reasons

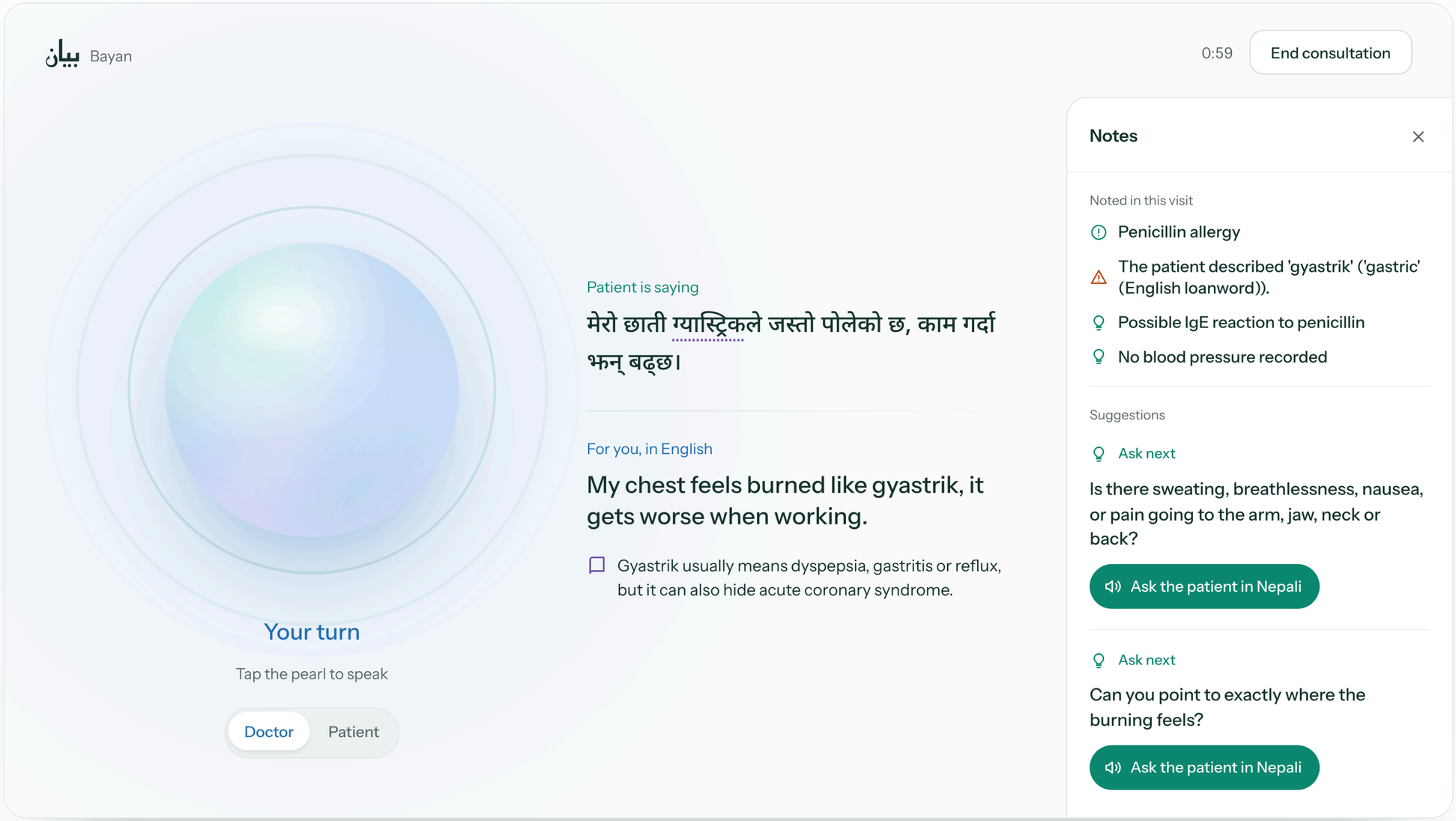
Keeps a running case summary and reviews it after every turn.

Flags

Allergies, doses, interactions, missed workup, plans that don’t fit the complaint, patient confusion, mood cues.

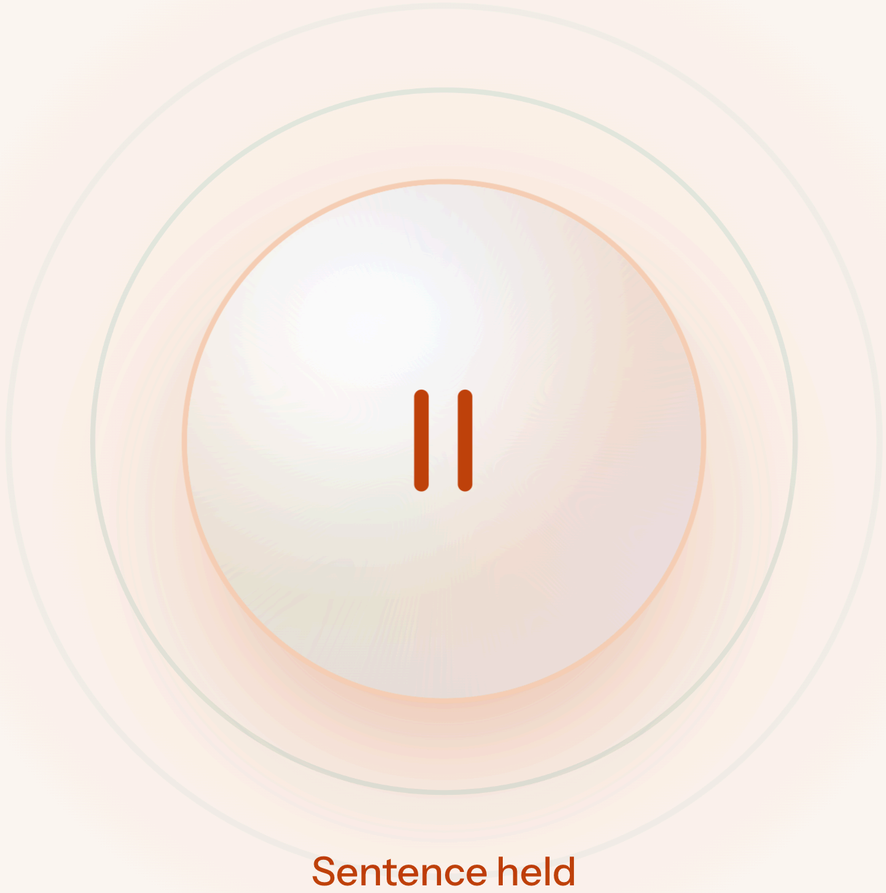
Suggests

One next question at a time, asked in the patient’s language with one tap.



Doctor and patient take turns tapping the pearl. Bayan translates, speaks, and checks.

Recorded on the live Space: scripted Nepali consultation, live translation and voices. Played at 1.5× speed.



Sentence held

The patient reported a penicillin allergy. Amoxicillin is a penicillin.

Patient: “When I was given a penicillin injection once, my body swelled.”

You: “Amoxicillin 500 three times a day, and you can go home.”

Say something else | Speak it anyway

Held before the patient hears it. Rule ALG-PEN-01, checked in about 3 ms, with no language model in the path. “Speak it anyway” is logged.

1

Care plan and doctor’s note



بيان Bayan

तपाईंको औषधि र हेरचाह निर्देशन

Your medicines and care instructions

07 Nov 2026

PDF डाउनलोड गर्नुहोस्
Download PDF

तपाईंको औषधि Your medicines

प्यारासिटामोल 500 मि.ग्रा. चक्की
Paracetamol 500 mg tablet
5 दिनसम्म खाना खाएपछि दिनको 3 पटक 2 चक्की खानुहोस्।
Take 2 tablets 3 times a day after food, for 5 days.

ओमेप्राजोल 20 मि.ग्रा. क्याप्सुल
Omeprazole 20 mg capsule
14 दिनसम्म हरेक बिहान खाना अघि 1 क्याप्सुल।
1 capsule every morning before breakfast, for 14 days.

दैनिक तालिका Daily schedule

उठ्ने बित्तिकै When you wake up	ओमेप्राजोल · 1 क्याप्सुल
बिहानको खाना Breakfast	प्यारासिटामोल · 2 चक्की

Phone page, via QR code



உங்கள் மருந்துகள் மற்றும் பராமரிப்பு வழிமுறைகள்

Your medicines and care instructions

07 Nov 2026

உங்கள் மருந்துகள்
Your medicines

அசித்ரோமைசின் 500 மி.கி. Azithromycin 500 mg
500 மி.கி. ஒரு முறை தினம் எடுத்துக்கொள்ளவும். 3 நாட்களுக்கு.
Take 500 mg once a day. For 3 days.

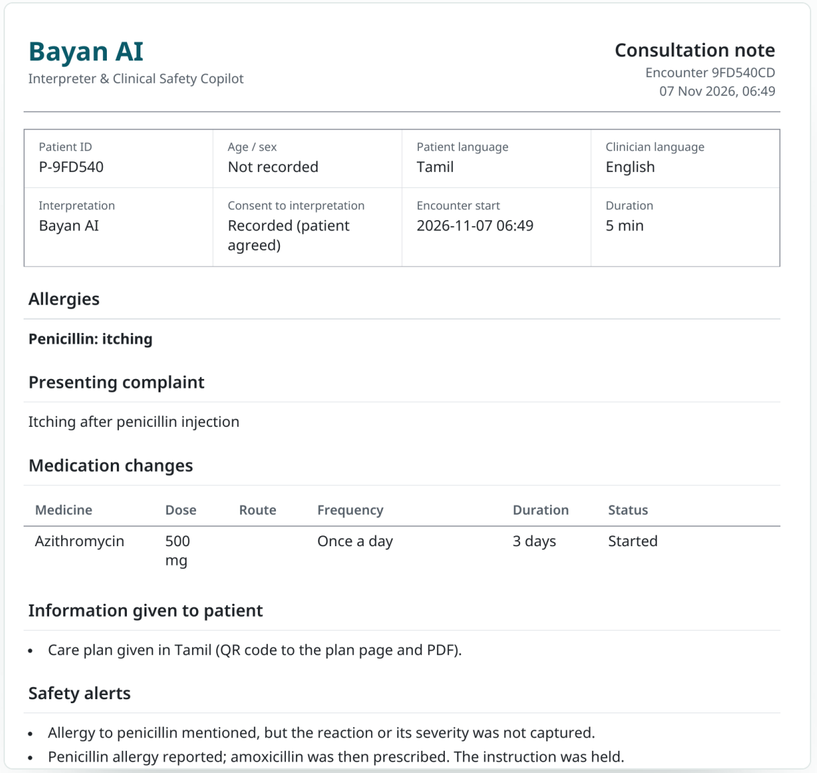
தினசரி அட்டவணை
Daily schedule

மருந்து Medicine	காலை உணவு Breakfast
அசித்ரோமைசின் Azithromycin	**500 மி.கி.** 500 mg

எடுக்க வேண்டாம்
Do not take

- பெனிசிலின் மருந்துகள், உதாரணமாக அமோக்சிசிலின். நீங்கள்

Patient sheet in Tamil: penicillins under “Do not take”



Bayan AI
Interpreter & Clinical Safety Copilot

Consultation note
Encounter 9FD540CD
07 Nov 2026, 06:49

Patient ID	Age / sex	Patient language	Clinician language
P-9FD540	Not recorded	Tamil	English
Interpretation	**Consent to interpretation**	**Encounter start**	**Duration**
Bayan AI	Recorded (patient agreed)	2026-11-07 06:49	5 min

Allergies
Penicillin: itching

Presenting complaint
Itching after penicillin injection

Medication changes

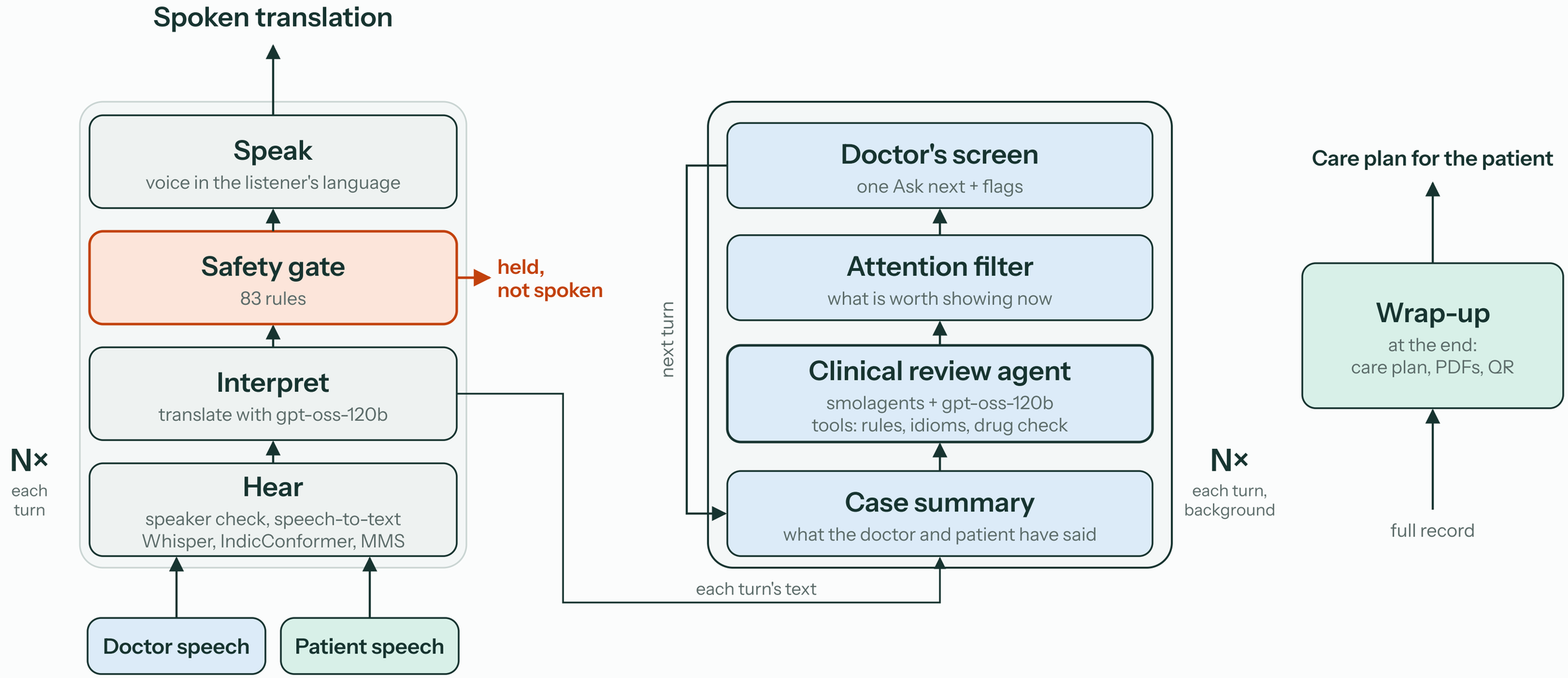
Medicine	Dose	Route	Frequency	Duration	Status
Azithromycin	500 mg		Once a day	3 days	Started

Information given to patient
- Care plan given in Tamil (QR code to the plan page and PDF).

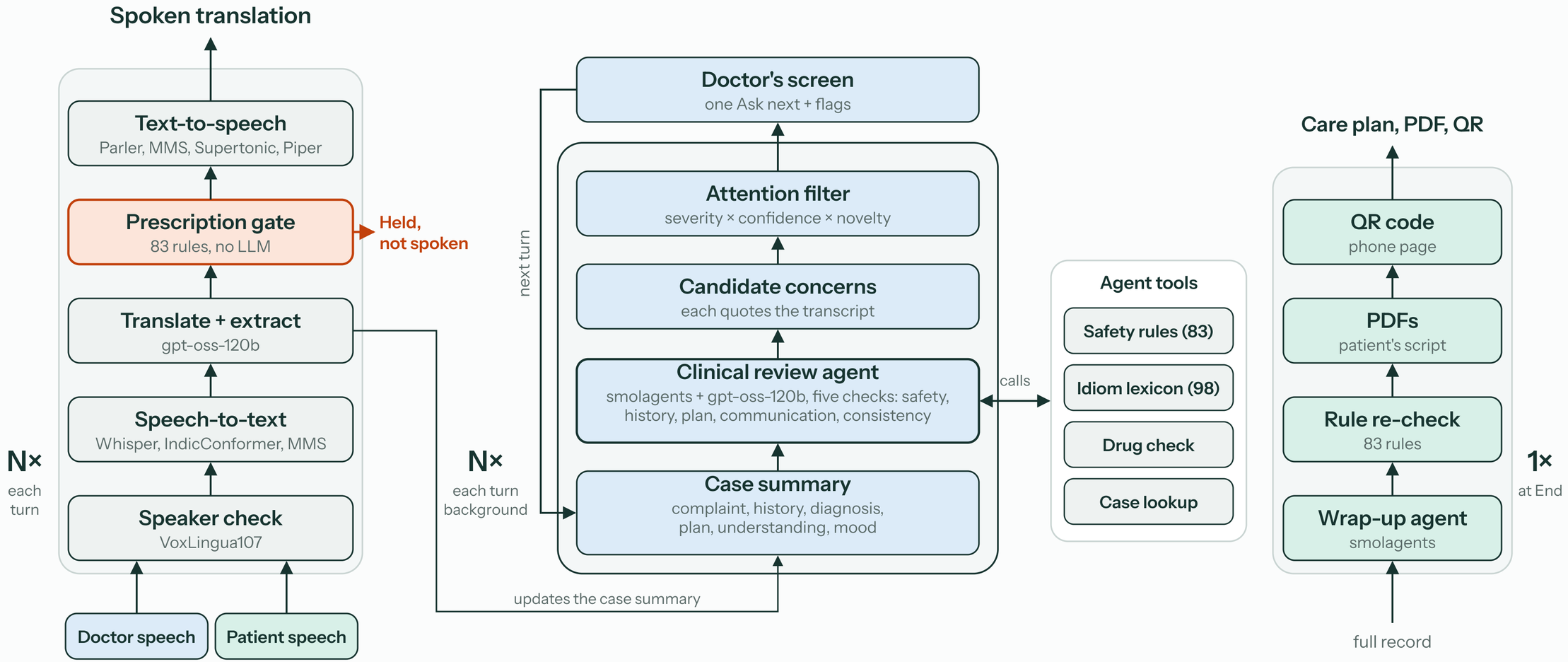
Safety alerts
- Allergy to penicillin mentioned, but the reaction or its severity was not captured.
- Penicillin allergy reported; amoxicillin was then prescribed. The instruction was held.

Doctor’s note: the held amoxicillin, on the record

Architecture (overview)



Architecture



Hugging Face stack

Bayan AI

CPU upgrade

dullgrey/team-qh-129-bayan-app

- gradio-app **gradio**
- huggingface **smolagents**

Bayan Voice Worker

NVIDIA L4

dullgrey/team-qh-129-bayan-voice

- openai **whisper-large-v3-turbo**
- facebook **mms-1b-all**
- ai4bharat **indic-parler-tts**
- Supertone **supertonic-3**
- ai4bharat **indic-conformer-600m-multilingual**
- speechbrain **lang-id-voxlingua107-ecapa**
- facebook **mms-tts-tgl, mms-tts-swh**
- rhasspy **piper-voices**

Inference Providers

Cerebras, Groq

router.huggingface.co

- openai **gpt-oss-120b**

Simulation results

Scripted consultations run through the real engine as text, with gpt-oss-120b. Results vary between runs.

Safety rules

38 consultations, 14 languages

99% of 727 checks passed

0 of 44 false-alarm traps tripped

100% of 477 PDF checks: native script, held drug never on the patient sheet

Clinical review

16 consultations, 9 languages: 7 with a plan that doesn't fit, 6 quiet visits, 3 repeats of rule cases

8 of 12 expected concerns flagged

0 false caution alerts on the 6 quiet visits

30 of 30 rule checks still passed, 0 of 22 traps tripped

Translation and gate: 0.9 s per turn. Agent card: about 7 s after the turn. LLM cost: about $0.014 per consultation.

First customers in Qatar

1.46 M

visits to the four QRCS workers' health centres in 2025

1.56 M

HMC emergency visits in 2025, adult and paediatric

5.17 M

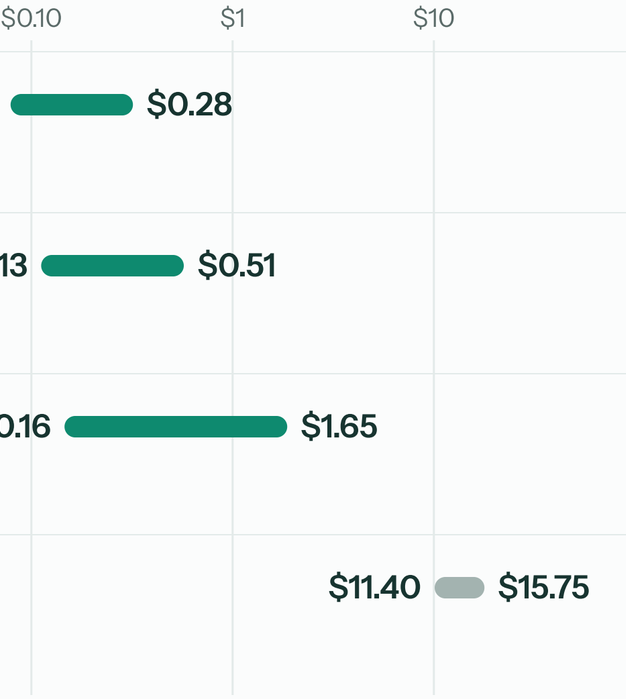
visits to PHCC's 31 health centres in 2024

Pricing to test: QAR 2–3 per consultation. A phone interpreter costs about QAR 40–60 per visit.

Deployment and cost

All models are open-weight, so the system can also run on a hospital's own servers.

Where it runs	Up-front hardware	Per clinic-month	Per consultation, log scale
Online on Hugging Face Voice on one NVIDIA L4 Space ($0.80/h), app on CPU ($0.03/h), gpt-oss-120b per token on Cerebras	None	$107 if 4 clinics share $339 alone	**$0.09** – **$0.28**
Clinic box 2× NVIDIA L4, about $2,800 each: all voice models plus gpt-oss-20b, on site	$14.6k QAR 53k	$607 alone $152 each if 4 share	**$0.13** – **$0.51**
Hospital, in-country 1× H100 80 GB, about $27.5k, runs gpt-oss-120b (MXFP4, one GPU); 1× L40S, about $8.5k, for voice	$48k QAR 175k	$1,980 per server $198 per clinic across 10	**$0.16** – **$1.65**
Phone interpreter CyraCom or Propio $0.76/min, LanguageLine $1.05/min; 15 minutes a visit	None	$13,680–18,900	**$11.40** – **$15.75**



Comparison

	Nepali, Malayalam, Bengali in the room	Checks allergy, dose, missed workup	Holds a sentence before the patient hears it	Can run in-country	Cost per visit
Bayan	**Yes, 37 languages**	**Yes, 83 cited rules**	**Yes**	**Yes, open weights**	**$0.09–0.28**
Phone and video interpreters	Yes, remote	Human judgement	No	No	$11–74
Google, Microsoft Translate	Text; speech unverified	No	No	No	Free
AI medical interpreters	Claimed 40–150+	Translation checks only	No	Some (Mabel)	Not public
Ambient AI scribes	No, English notes	No, documentation	No	No	$150–600 per doctor-month

Next steps

A pilot at one QRCS workers' health centre

Clinicians to review our 83 rules and 98 idioms

Native speakers to validate each language

Team QH-129
huggingface.co/spaces/dullgrey/team-qh-129-bayan-app